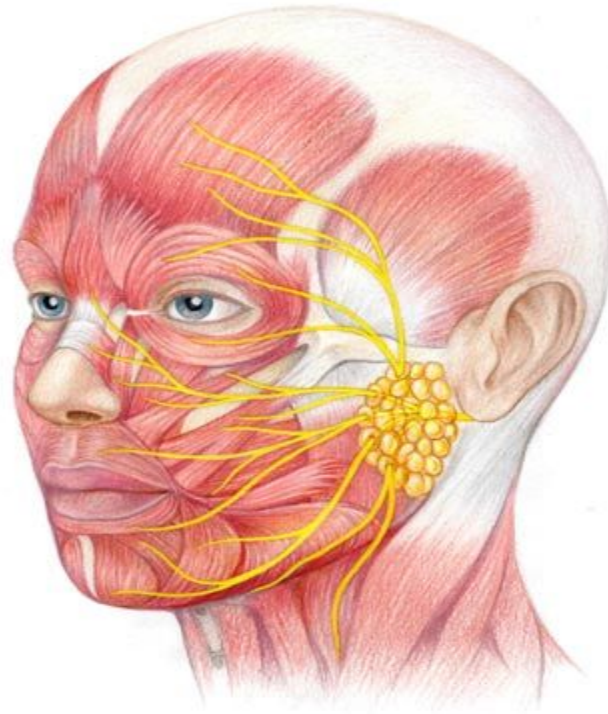


The Facial Nerve (VII)



Learning objectives:

At the end of the lecture, the students should be able to:

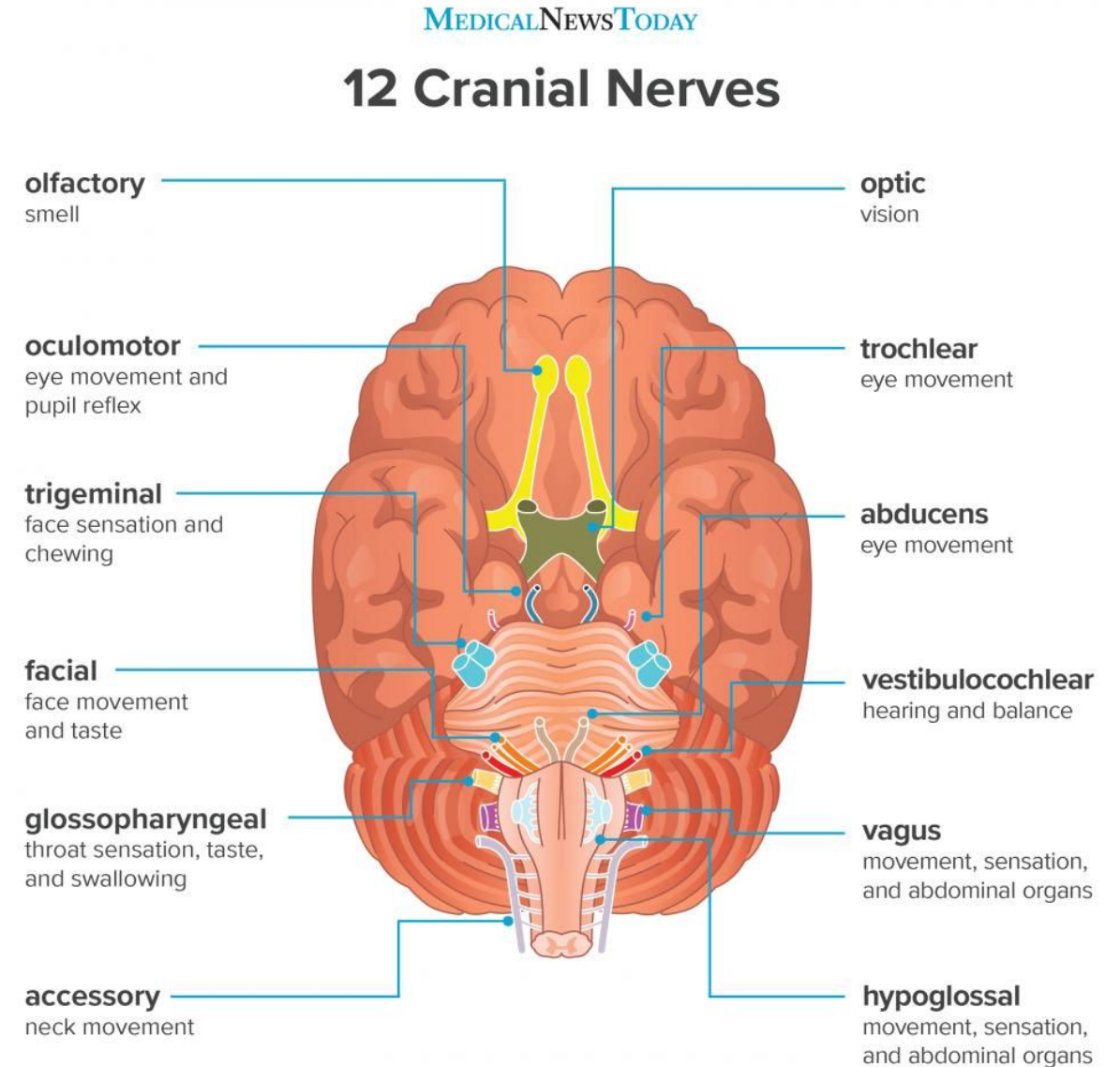
- **Describe the basic introduction of the facial nerve.**
- **Enlist the nuclei related to the nerve.**
- **Explain the course of the nerve.**
- **Enlist the branches of facial nerve and explain their distribution.**
- **Define facial palsy, the supranuclear & the infranuclear lesions.**

Introduction:

- It is the seventh cranial nerve.
- It is mixed nerve.
- It is a nerve of second pharyngeal arch.

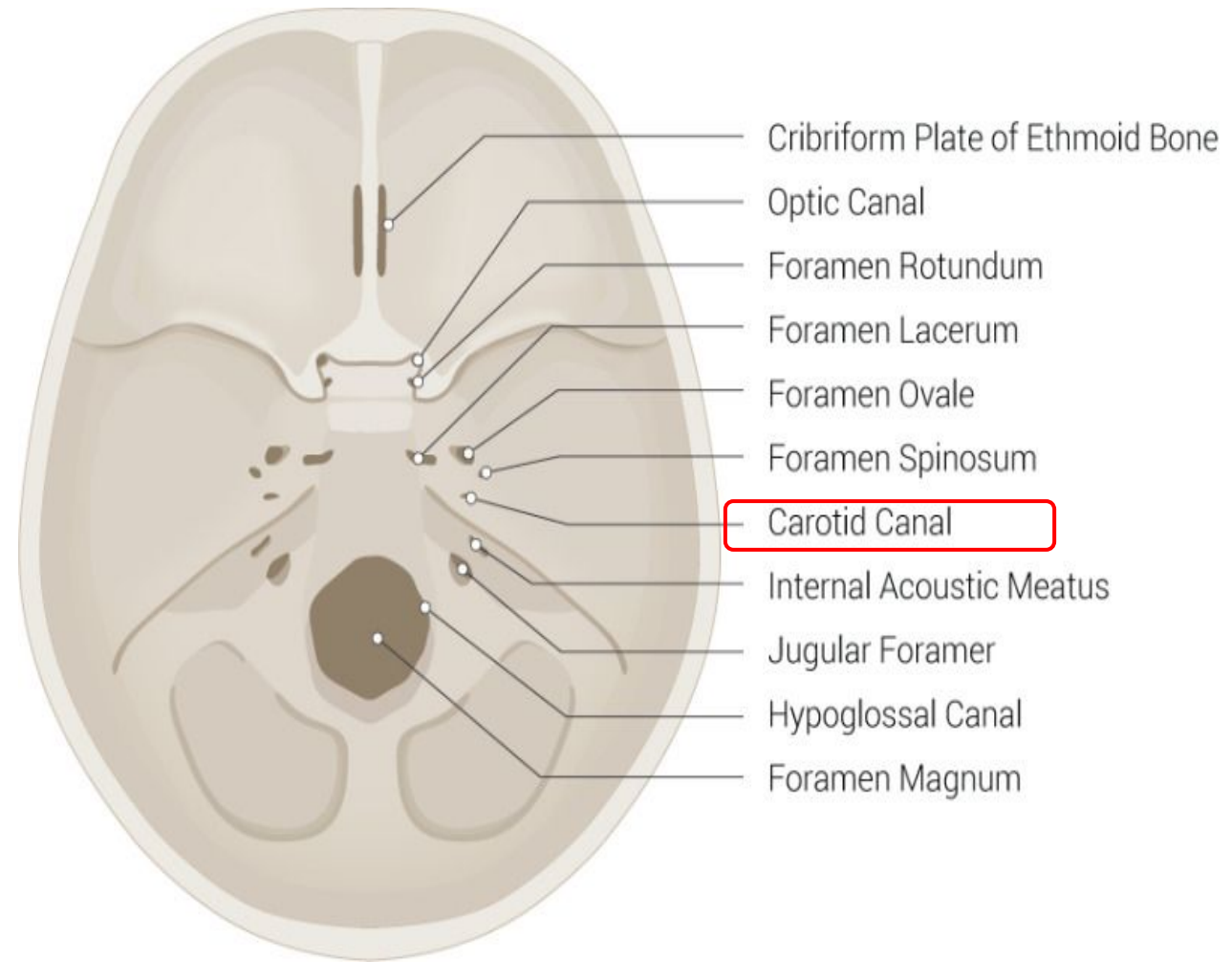
Nuclei:

- 1) Motor nucleus
- 2) Superior salivatory nucleus (parasympathetic)
- 3) Lacratory nucleus (parasympathetic)
- 4) Nucleus of tractus solitarius

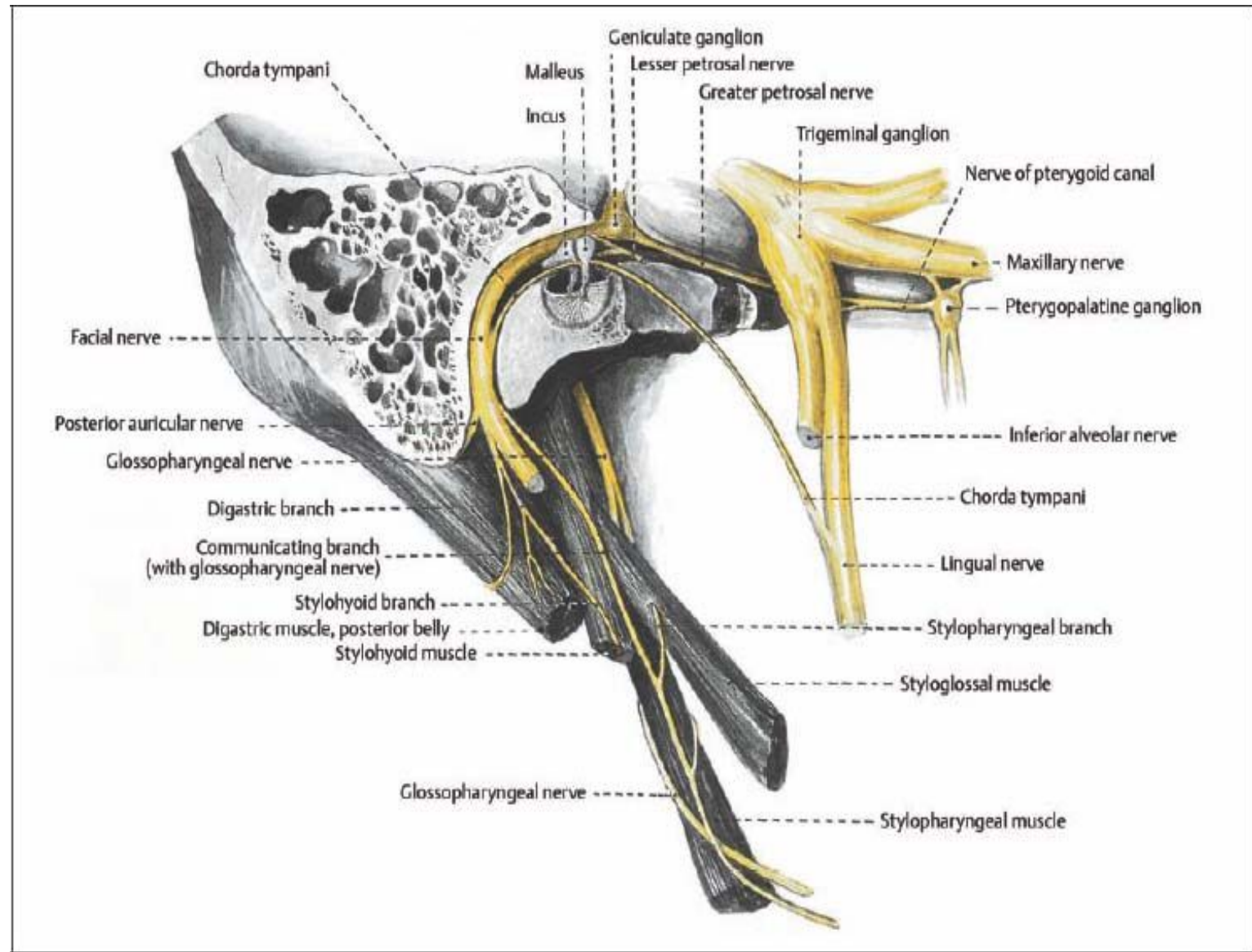


Course:

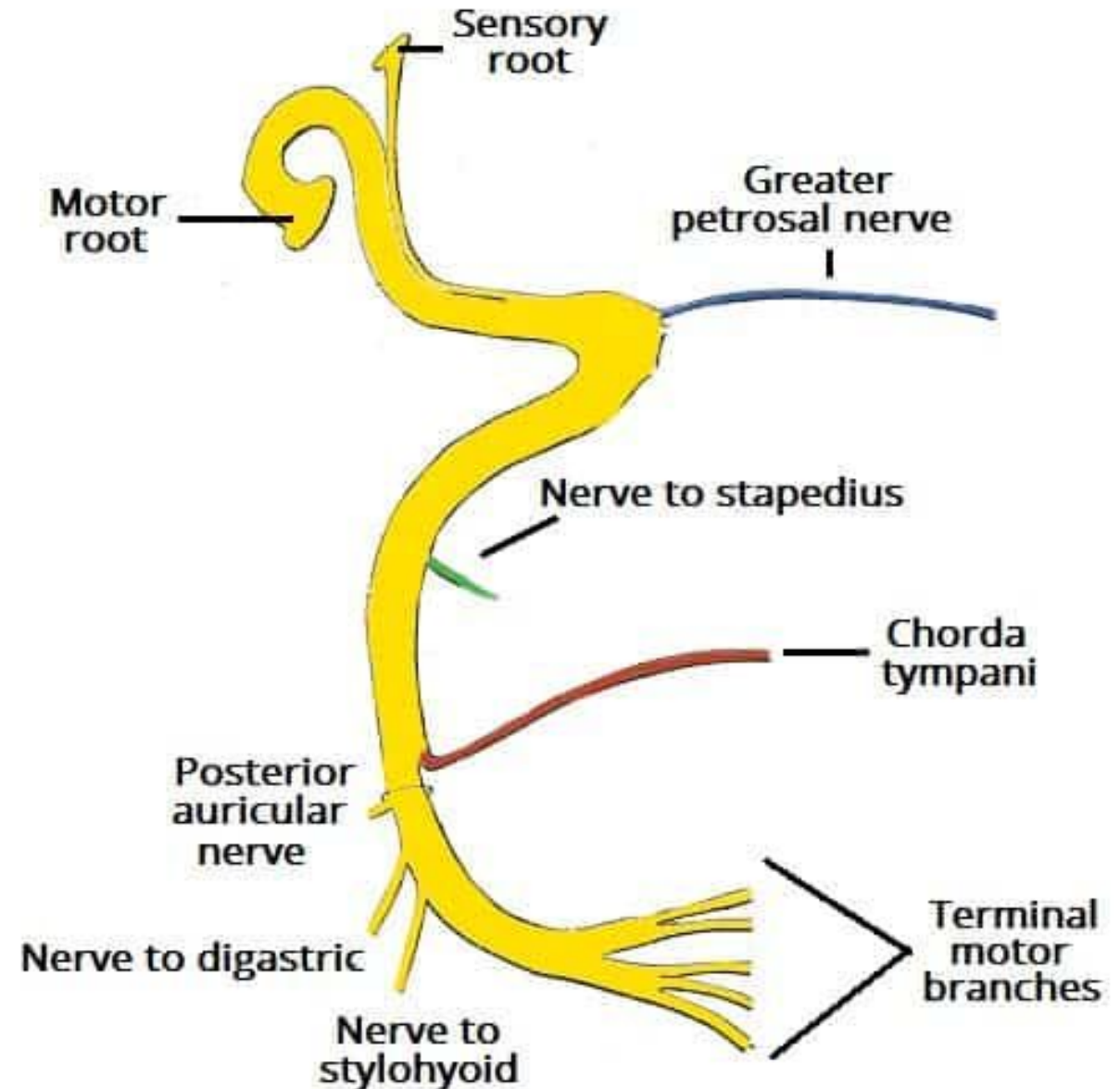
- Facial nerve **arises from pontomedullary junction** medial to eighth cranial nerve.
- It has **two roots**, the **motor** root **and** the **sensory** root called nervus intermedius.
- The two roots enter the internal acoustic meatus and unite to form a single trunk.
- It lies in the facial canal within petrous temporal bone.



- It has three parts and two bends within the canal.
- The first part lies laterally above vestibule, the second part is related to medial wall of middle ear above promontory and the third part lies behind promontory.
- The **first bend** between first and second parts lies superior to promontory, it is called **genu**.
- The **geniculate ganglion** lies on genu.



- The facial nerve **exits the skull through stylomastoid foramen.**
- It crosses the base of styloid process and enters the parotid gland through posteromedial surface.
- Behind the neck of mandible, in the parotid gland it divides into terminal branches.



Branches:

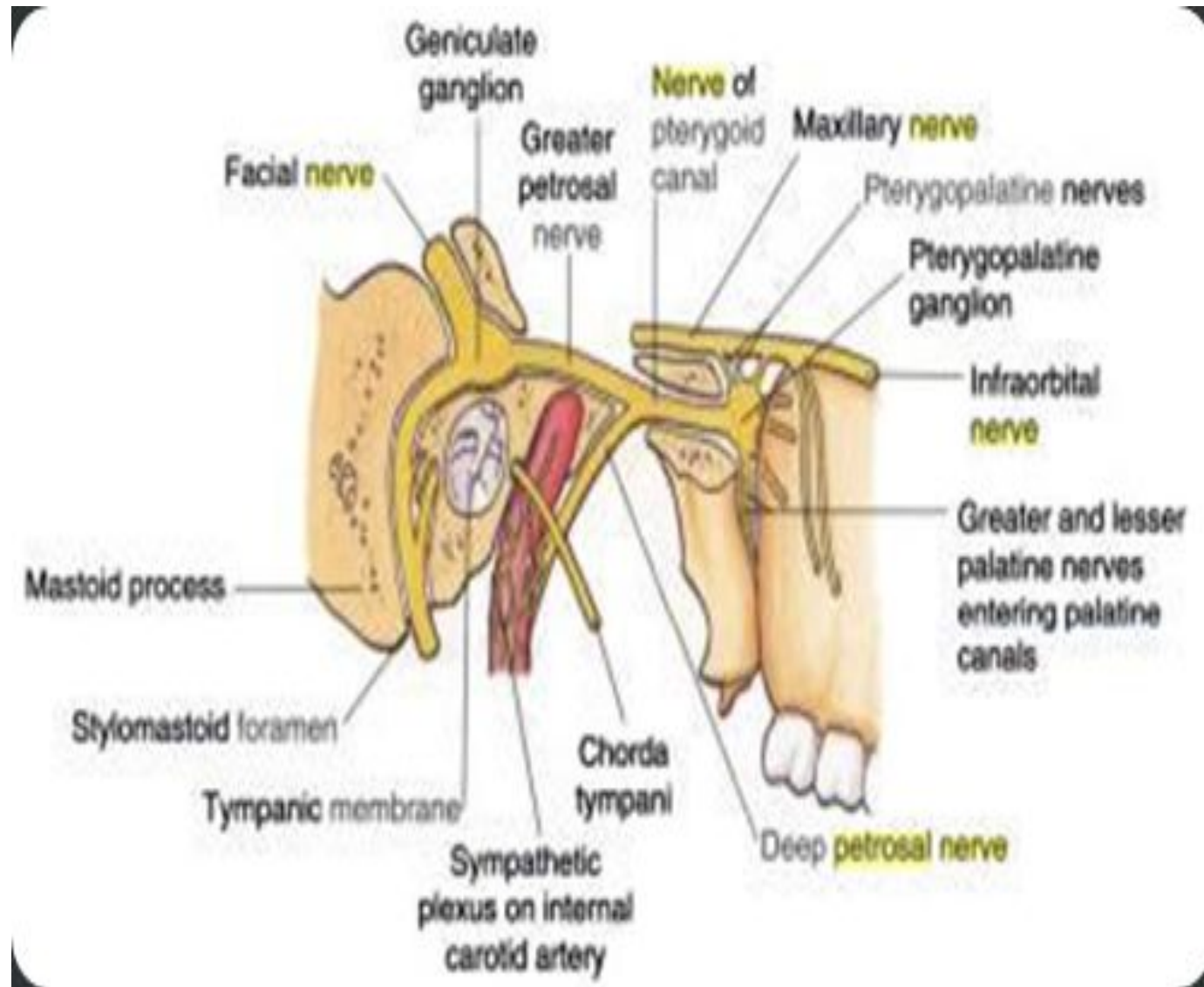
In the facial canal (petrous temporal bone):

1. Greater petrosal nerve:

- Arises from geniculate ganglion.
- Carries gustatory fibers to the palate.
- Also carries parasympathetic fibers to the pterygopalatine ganglion via nerve of pterygoid canal.
- Joins the deep petrosal nerve (sympathetic) to form the nerve of pterygoid canal.
- Postganglionic parasympathetic fibers supply the lacrimal gland, mucosal glands of nose, palate & pharynx.

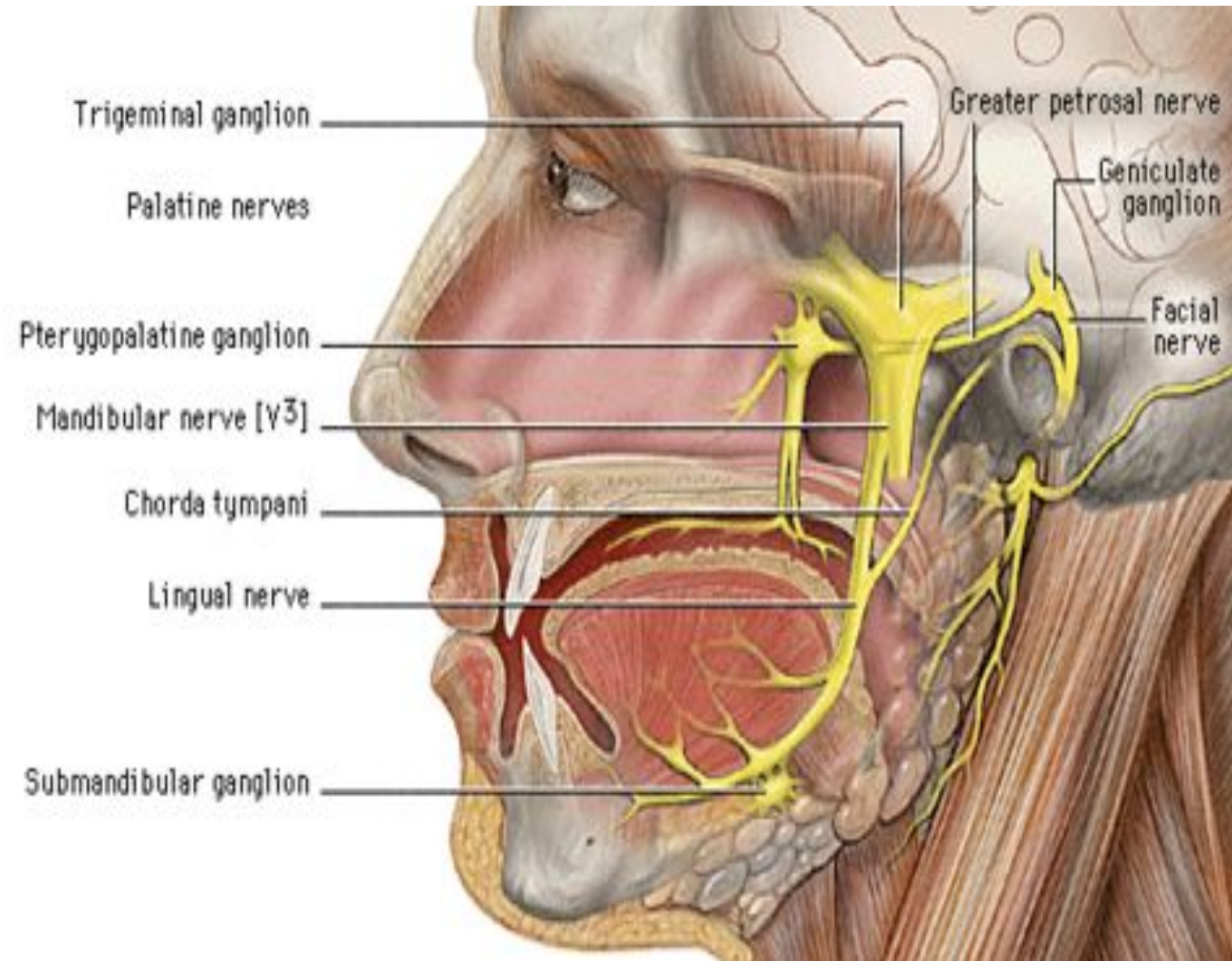
2. Nerve to stapedius:

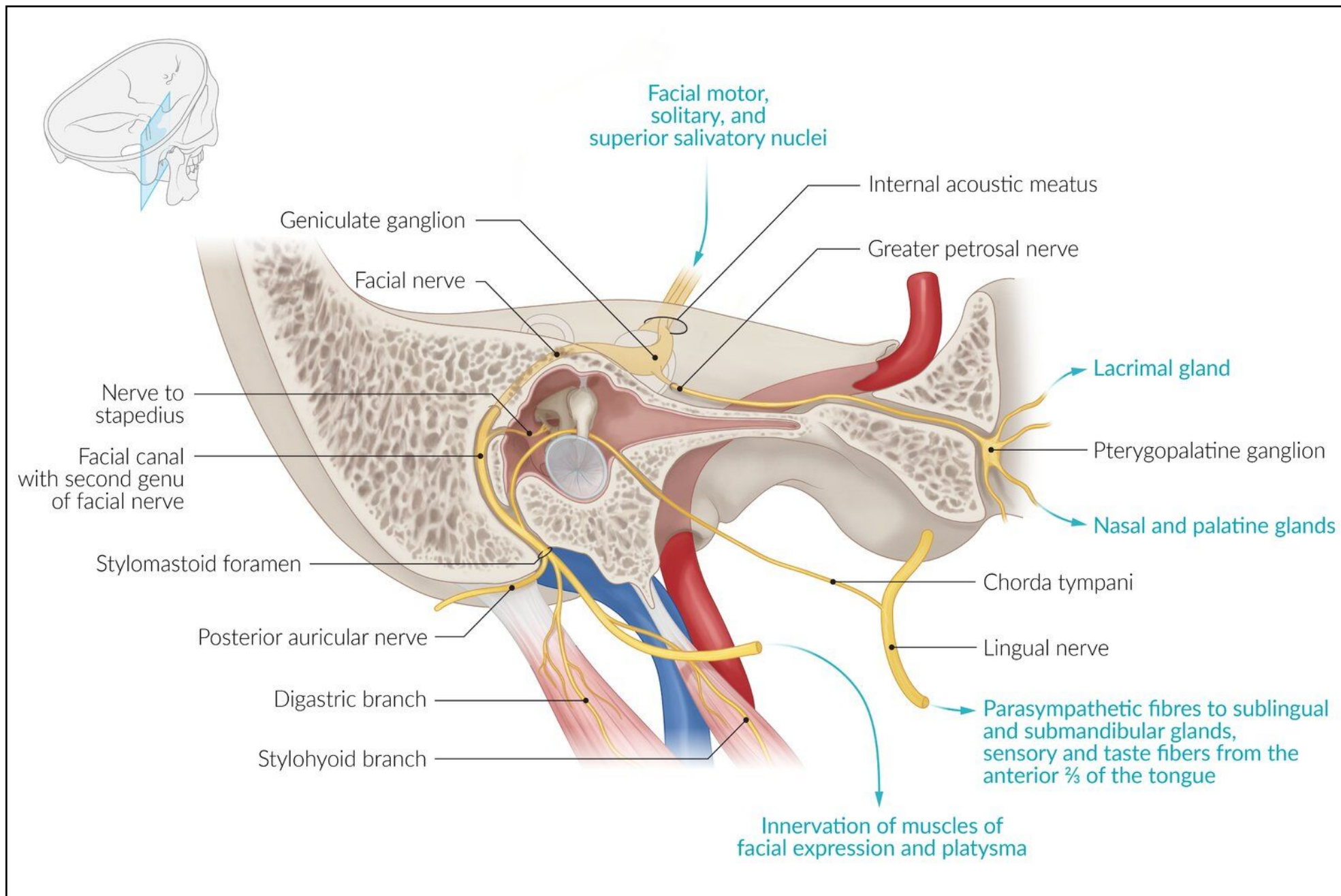
- Supplies the stapedius muscle.
- Its paralysis may lead to **hyperacusis**.



3. Chorda tympani:

- It arises in the facial canal and enters the middle ear.
- Exits the middle ear & lies in the infratemporal fossa.
- In the infratemporal fossa, it joins the lingual nerve.
- It carries **taste sensations from anterior 2/3rds of the tongue.**
- Also carries preganglionic **secretomotor fibers to the submandibular ganglion for submandibular & sublingual salivary glands.**



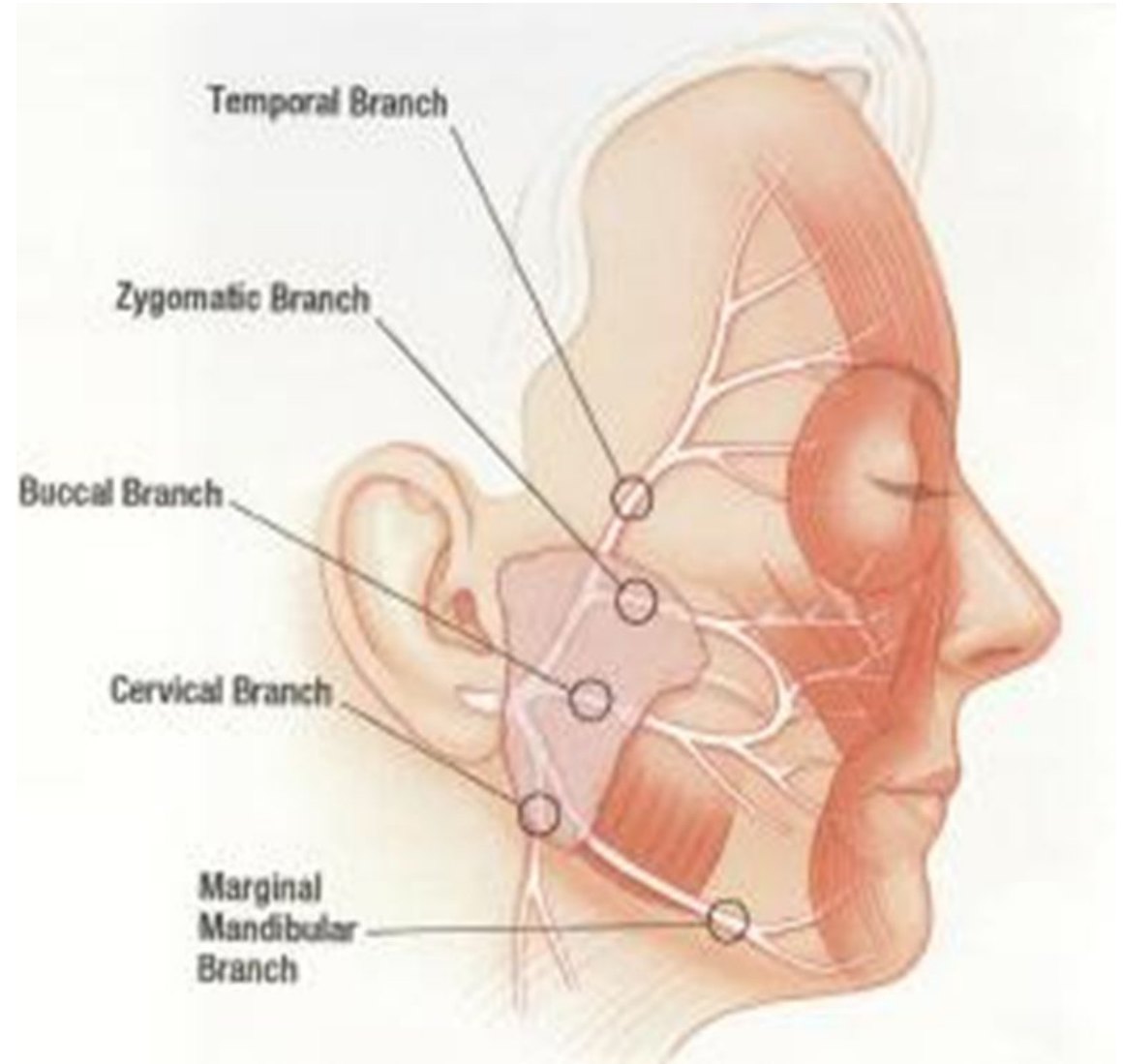


After emerging from stylomastoid foramen:

- 1) Posterior auricular nerve**
- 2) Digastric nerve** (to posterior belly of digastric)
- 3) Stylohyoid nerve**

Terminal branches:

- 1) Temporal**
- 2) Zygomatic**
- 3) Buccal**
- 4) Marginal mandibular**
- 5) Cervical**



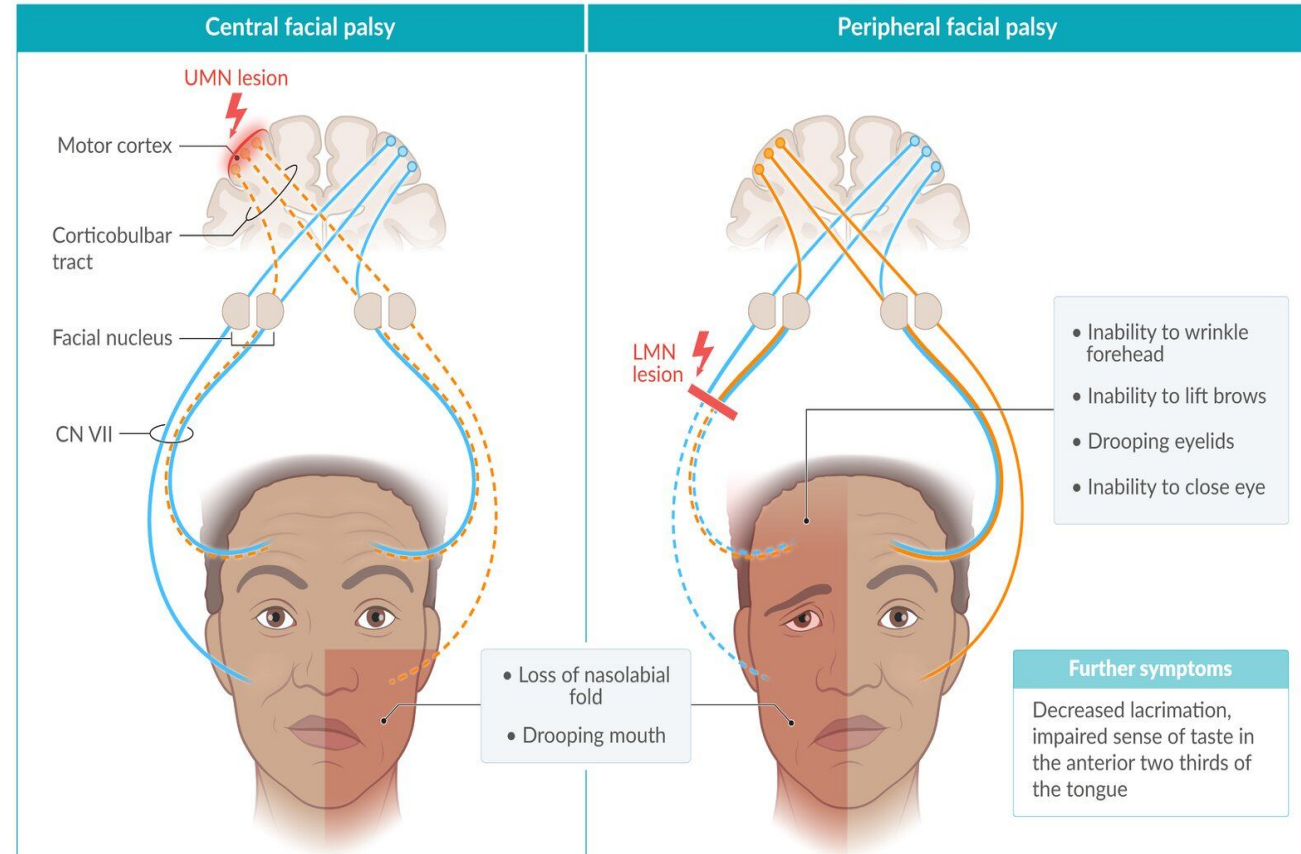
Facial nerve palsy:

Supranuclear lesion:

- It is usually a part of hemiplegia.
- Only lower part of opposite side of face is paralysed.
- Upper part with frontalis & orbicularis oculi escapes due to its bilateral representation in cerebral cortex.

Infranuclear lesion:

- In this, whole of face of the same side gets paralysed.



References:

- **BD CHAURASIA'S HUMAN ANATOMY (Head, Neck, & Brain)-Vol. 3**

4th edition

Chapter 8

Pages # 138-141